Optimizing Real Estate Pricing through Data Analytics and Machine Learning

Project Report

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1. Business Problem:

The main business problem addressed is the inefficiency and unreliability of quantitative insights in conventional real estate evaluation. Traditional approaches to valuation often depend on judgments or straightforward comparable sales analysis, which fail to sufficiently capture the complex, non - linear relationships among various property characteristics. This inadequacy leads results in less than ideal decisions for all stake holders involved in real estate; agents find it challenging to establish listing prices, investors overlook undervalued properties and buyers face the risk of overpaying. To address this challenge, this project aims to implement a data-driven predictive framework for property valuation and market segmentation.

2. Analytical Questions:

- How accurately can machine learning models predict housing prices using property features?
- Which specific characteristics of properties such as size, construction quality, condition, and location have the most significant impact on price variations and how do these attributes influence final valuations in a no n linear way?
- Can a classification model reliably identify high-value homes to support better pricing, marketing, and investment decisions?

3. Main Objectives:

- Build and test models that can predict house prices and classify properties into market tiers.
- Compare their performance using relevant metrics to find which model gives the most reliable results.
- Identify which features drive prices the most and turn those findings into insights for pricing, renovation, and investment planning.

4. Hypotheses:

- **H1:** Homes with more living space and higher construction grades will sell for higher prices.
- **H2:** Homes that were recently renovated will have a higher value relative to similar homes that were unrenovated.
- **H3:** Models like Random Forest and Gradient Boosting will perform better than linear models because these models can pick up on complex relationships between features.

5. Data Description and Preparation:

5.1 Data Source:

The data used in this project comes from the King County House Sales Research Dataset, publicly posted on Kaggle (which is comprised of data collected by the King County Assessor's Office in Seattle, Washington, USA) with detailed records of residential housing sales that took place between May 2014 and May 2015.

The dataset contains 21,613 observations and is comprised of 21 different variables that contain structural, locational and transactional information for residential homes. The data can be sourced from the following location:
Kaggle – "House Sales in King County, USA"
https://www.kaggle.com/datasets/harlfoxem/housesalesprediction

This dataset is widely used for property price modelling because it combines both quantitative attributes (e.g., living area, number of bedrooms, lot size) and qualitative characteristics (e.g., condition, grade, waterfront view, location coordinates).

5.2 Variable Overview:

The table below summarizes the key variables included in the analysis and their descriptions:

Variables	Descriptions
• **id:**	Unique identifier for each property.
• **date:**	Date when the property was sold.
• **price:**	Sale price of the house (target variable for regression).
• **bedrooms:**	Number of bedrooms.
• **bathrooms:**	Number of bathrooms (can include partial bathrooms).
• **sqft_living:**	Living area size (in square feet).
• **sqft_lot:**	Total lot size (in square feet).
• **floors:**	Number of floors (levels) in the house.
• **waterfront:**	Binary indicator (1 if property has a waterfront view, 0 otherwise).
• **view:**	Quality of view from the property (ordinal scale: 0–4).
• **condition:**	Overall condition of the house (scale: 1–5).
• **grade:**	Construction quality and design (scale: 1–13).
• **sqft_above:**	Square footage of the interior space above ground level.
• **sqft_basement:**	Square footage of the basement area.

• **yr_built:**	Year when the house was originally constructed.
• **yr_renovated:**	Year of last renovation (0 if never renovated).
• **zipcode:**	ZIP code of the property's location.
• **lat, long:**	Geographic coordinates (latitude and longitude).
• **sqft_living15:**	Average living area of 15 nearest neighbors.
• **sqft_lot15:**	Average lot area of 15 nearest neighbors

6. Primary Overview of the Dataset:

```python
# Summary statistics (mean, median, std) for numeric columns

numeric_cols = df.select_dtypes(include=['number']).columns.tolist()

summary = df[numeric_cols].describe().T  # contains mean, std, min, 25%, 50%, 75%, max
summary['median'] = df[numeric_cols].median()
summary = summary[['mean', 'median', 'std', 'min', '25%', '50%', '75%', 'max', 'count']]
summary.style.format("{:.3f}")
```

	mean	median	std	min	25%	50%	75%	max	count
id	4580306147.710	3904930410.000	2876569751.889	1000102.000	2123049194.000	3904930410.000	7308900445.000	9900000190.000	21613.000
price	540088.577	450000.000	367126.825	75000.000	321950.000	450000.000	645000.000	7700000.000	21613.000
bedrooms	3.371	3.000	0.930	0.000	3.000	3.000	4.000	33.000	21613.000
bathrooms	2.115	2.250	0.770	0.000	1.750	2.250	2.500	8.000	21613.000
sqft_living	2079.900	1910.000	918.441	290.000	1427.000	1910.000	2550.000	13540.000	21613.000
sqft_lot	15106.968	7618.000	41420.512	520.000	5040.000	7618.000	10688.000	1651359.000	21613.000
floors	1.494	1.500	0.540	1.000	1.000	1.500	2.000	3.500	21613.000
waterfront	0.008	0.000	0.087	0.000	0.000	0.000	0.000	1.000	21613.000
view	0.234	0.000	0.766	0.000	0.000	0.000	0.000	4.000	21613.000
condition	3.409	3.000	0.651	1.000	3.000	3.000	4.000	5.000	21613.000
grade	7.657	7.000	1.175	1.000	7.000	7.000	8.000	13.000	21613.000
sqft_above	1788.391	1560.000	828.091	290.000	1190.000	1560.000	2210.000	9410.000	21613.000
sqft_basement	291.509	0.000	442.575	0.000	0.000	0.000	560.000	4820.000	21613.000
yr_built	1971.005	1975.000	29.373	1900.000	1951.000	1975.000	1997.000	2015.000	21613.000
yr_renovated	84.402	0.000	401.679	0.000	0.000	0.000	0.000	2015.000	21613.000
zipcode	98077.940	98065.000	53.505	98001.000	98033.000	98065.000	98118.000	98199.000	21613.000
lat	47.560	47.572	0.139	47.156	47.471	47.572	47.678	47.778	21613.000
long	-122.214	-122.230	0.141	-122.519	-122.328	-122.230	-122.125	-121.315	21613.000
sqft_living15	1986.552	1840.000	685.391	399.000	1490.000	1840.000	2360.000	6210.000	21613.000
sqft_lot15	12768.456	7620.000	27304.180	651.000	5100.000	7620.000	10083.000	871200.000	21613.000

6.1 Summary Statistics and Interpretation:

This is the first step in EDA, giving a statistical outlook of the dataset before any modelling. It helps in identifying outliers, skewness, and data inconsistencies, and also reveals some variance between property characteristics and pricing. From the summary output, several facts come to light:

- **Price**:

  The mean home price is around $540,088 USD with a median of $450,000 USD, suggesting right-skewed distribution: a few high-value homes ran the mean higher than the median. This

indicates there are luxury or waterfront properties that could have a significant economic impact, if not vast, on pricing models.

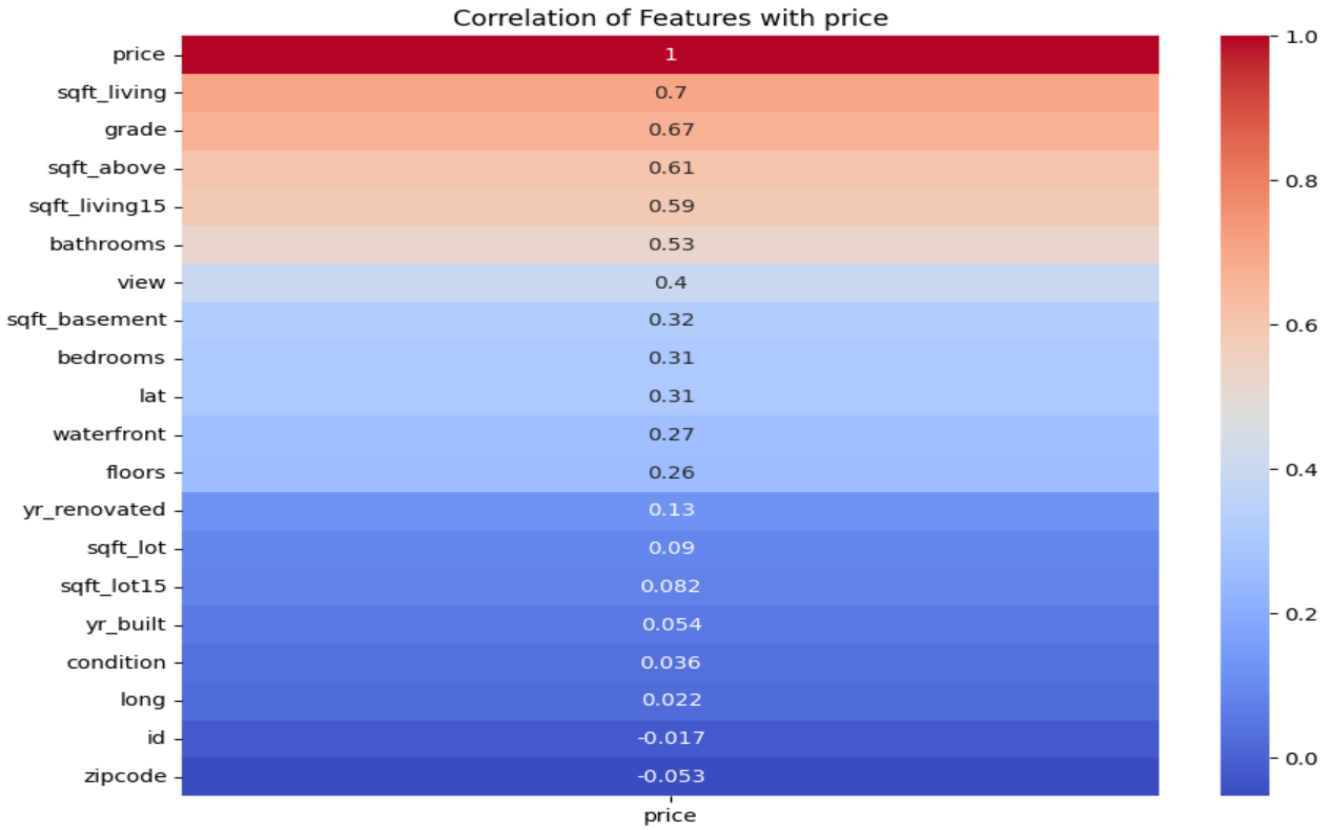
- **Living Area (sqft_living)**:
  The following indicates the mean living space is about 2,079 sq. ft., and has a median of 1,910 sq. ft. This moderate difference suggests that most homes must fall within a fairly tight range to support the notion of consistency in size across the dataset possibly due to zoning or architectural norms.
- **Lot Size (sqft_lot)**:
  The average lot size, 15,107 sq. ft., is so much greater than the median, 7,618 sq. ft., indicating that there are only a few properties that are very large-rural or premium estates-which inflate the mean. An extreme maximum value of over 1.6 million sq. ft. confirms the presence of strong outliers.
- **Bedrooms and Bathrooms**:
  The average number of bedrooms is 3.37, and bathrooms 2.12-a typical suburban family housing. The maximum, though, is 33 bedrooms and 8 bathrooms, highly suggestive that there may be a few really extreme listings, probably mansions or multi-unit dwellings.
- **Waterfront and View**:
  Means as low as 0.008 and 0.234 respectively suggest that few properties have waterfront access or premium views, and these features are highly differentiated and will likely contribute to price differences.
- **Year Built and Renovated**:
  The mean year built is about 1971, and a near-zero median of yr_renovated indicates that renovations are infrequent. However, the wide standard deviation of renovations-401.679-points to a number of recent major upgrades that may be key price drivers.
- **Grade and Condition**:
  With an average grade of 7.65 out of 13 and condition of 3.4 out of 5, most homes are pretty well maintained and above average quality. This variable could be positively correlated with price
- **Geographic Coordinates (lat, long)**:
  The density of latitude and longitude suggests a bounded geographic dataset and is likely to be of a metropolitan or regional market in nature, such as King County, Washington State.

6.2 Correlation Heatmap:

```python
# Correlation of Features With Target

import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10,8))
corr = df.corr(numeric_only=True)
sns.heatmap(corr[['price']].sort_values(by='price', ascending=False), annot=True, cmap='coolwarm')
plt.title('Correlation of Features with price')
plt.show()
```



A basic heatmap is generated to quantify the strength and direction of the relationships between the target variable (price) and all other numeric features in this dataset. The analysis shows that the living area has the highest dependence on the price of a house (0.70), implying that bigger houses attract higher market prices**. Grade (0.67)** and **above-ground living space (sqft_above, 0.61)** also have strong positive dependants, insinuating that construction quality and useful area is key in determining house prices. A moderate dependency is realized in both bathrooms, 0.53, and view, 0.40, implying design and aesthetics are major contributors but not decisive.

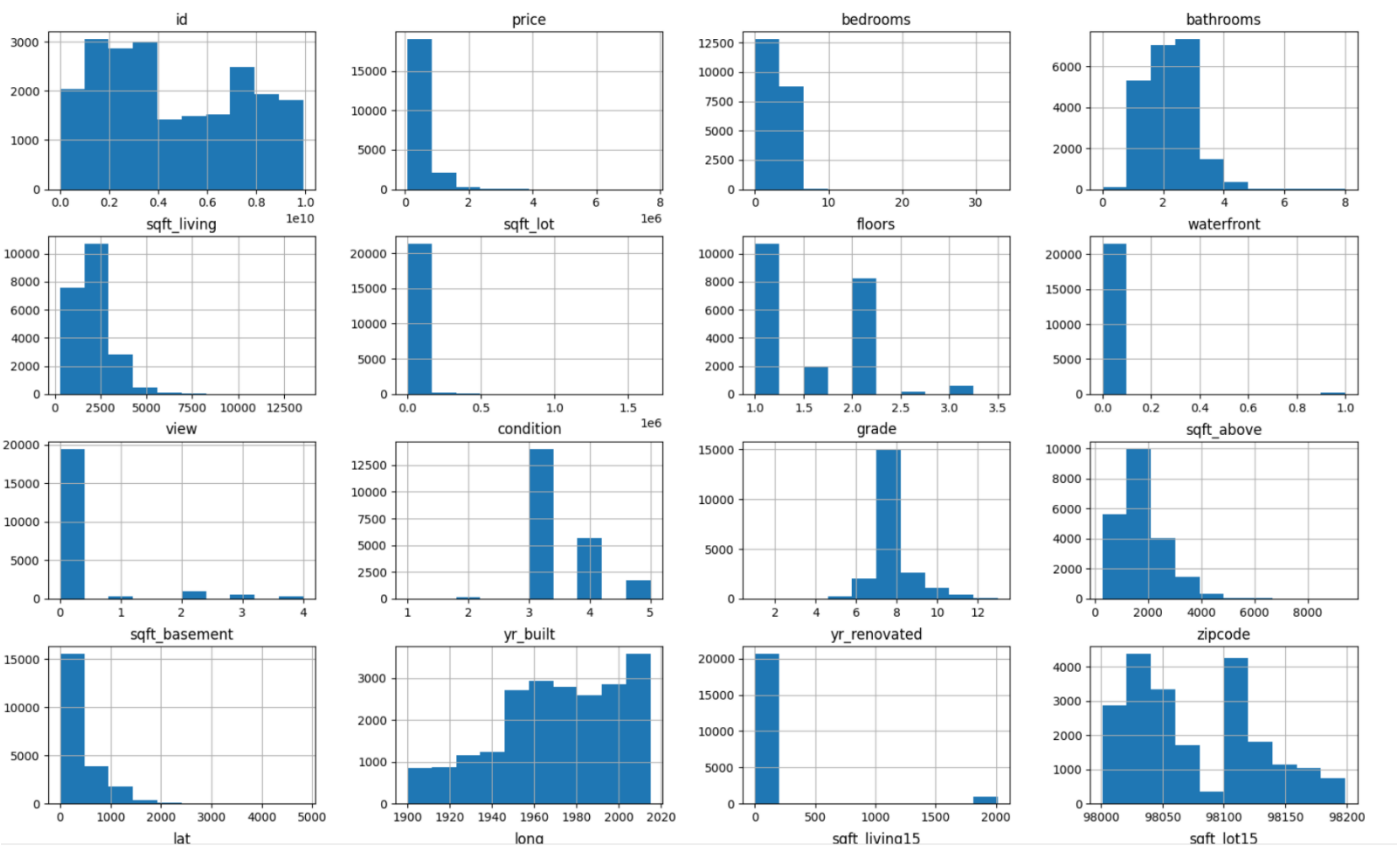
Features like lot size, year built, and year renovated exhibit relatively weak correlations and therefore have less linear effect on price in this data. Similarly, the almost zero values of longitude, ID, and zip code reassure their status regarding irrelevant features for prediction.

6.3 Histograms of Variables:

```python
# Initial Visualisation of Raw Data

numeric_col.hist(figsize=(20,15))

plt.show()
```



This visualization gives the first snapshot of distributional overview of all numeric variables in the dataset, displaying their histograms. Among key variables, such as price, living area-sqft_living, lot size-sqft_lot, and basement area, most of the distributions are right-skewed; thus, most of the observations are concentrated on the left side with a long tail on the right side. This is evidence that there are a few very expensive or large properties, confirming market heterogeneity.

Features such as grade and condition are normally distributed around the middle value, suggesting that most homes are of average to above-average quality. In contrast, the trend in year built is toward 2000, reflecting a modernizing housing stock, while year renovated reflects that most properties have never been renovated, although a few have been recently.

Categorical-like numeric features of floors, view and waterfront are highly imbalanced - most houses don't have premium attributes, like waterfront or even view. This distribution is helpful to explain the weakness of these variables correlations with the price, described earlier.

6.4 Skewness of Numeric Features:

```python
# Checking Skewness of Numerical Features
numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns

skewness = df[numeric_cols].skew().sort_values(ascending=False)
print("\n--- Skewness of Numerical Features ---")
print(skewness)
```

```
--- Skewness of Numerical Features ---
sqft_lot         13.060019
waterfront       11.385108
sqft_lot15        9.506743
yr_renovated      4.549493
price             4.024080
view              3.395750
bedrooms          1.974028
sqft_basement     1.577965
sqft_living       1.471555
sqft_above        1.446664
sqft_living15     1.108181
condition         1.032805
long              0.885053
grade             0.771103
floors            0.616177
bathrooms         0.511108
zipcode           0.405661
id                0.243327
yr_built         -0.469805
lat              -0.485270
dtype: float64
```

The output indicates that several variables are highly right-skewed. The sqft_lot (13.06), waterfront (11.39), and sqft_lot15 (9.51) are very highly skewed, which in turn suggests the existence of a few properties with exceptionally large land areas or highly valued places that differ significantly from the majority. The price variable also depicts significant right skewness at 4.02, reassuring that very highly valued properties are very rare but have strong influence on the overall price distribution. Moderately skewed features include bedrooms, sqft_basement, sqft_living, and view; they have indicated the fact that most homes are averagely sized and featured, with the smaller segment high class or unique properties. Only yr_built and lat exhibit mild left skew, which usually is not a problem.

7. Data Wrangling:

7.1 Basic information about the dataset:

```python
# Basic info of Data

df.info()

```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21613 entries, 0 to 21612
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   id             21613 non-null  int64
 1   date           21613 non-null  object
 2   price          21613 non-null  float64
 3   bedrooms       21613 non-null  int64
 4   bathrooms      21613 non-null  float64
 5   sqft_living    21613 non-null  int64
 6   sqft_lot       21613 non-null  int64
 7   floors         21613 non-null  float64
 8   waterfront     21613 non-null  int64
 9   view           21613 non-null  int64
 10  condition      21613 non-null  int64
 11  grade          21613 non-null  int64
 12  sqft_above     21613 non-null  int64
 13  sqft_basement  21613 non-null  int64
 14  yr_built       21613 non-null  int64
 15  yr_renovated   21613 non-null  int64
 16  zipcode        21613 non-null  int64
 17  lat            21613 non-null  float64
 18  long           21613 non-null  float64
 19  sqft_living15  21613 non-null  int64
 20  sqft_lot15     21613 non-null  int64
dtypes: float64(5), int64(15), object(1)
memory usage: 3.5+ MB
```

The dataset consists of 21 variables and 21,613 observations, representing a sufficiently large and acceptable sample size for regression analysis. The data types included are 15 integer (int64), 5 floating-point (float64), and 1 object (date) variable types, suggesting that most of the features are numeric, which are suitable as inputs for statistical modelling and almost all machine learning algorithms, indeed, even the date column is stored as an object type, which would need to be processed into a datetime type for better understanding of the data.

7.2 Missing Value Checking:

```python
# Checking Missing Values

df.isnull().sum()
```

	e
id	0
date	0
price	0
bedrooms	0
bathrooms	0
sqft_living	0
sqft_lot	0
floors	0
waterfront	0
view	0
condition	0
grade	0
sqft_above	0
sqft_basement	0
yr_built	0
yr_renovated	0
zipcode	0
lat	0
long	0
sqft_living15	0
sqft_lot15	0

dtype: int64

No column for the dataset contains any missing values, meaning that every variable from the property characteristics of sqft_living and grade to locational attributes of lat and zipcode contains full data. This means very good quality of the data, hence consistent and reliable throughout.

```python
# Checking for Duplicates

df.duplicated().sum()
```

```
np.int64(0)
```

7.3 Duplicate Entry Check:

A result of 0 means that there are no duplicate entries in the dataset. Each entry in the dataset is a distinct property listing, which increases our confidence in the dataset's analysis-relevant quality and validity. Duplicate entries in real estate datasets could arise if the same property is listed more than once under one single property record with multiple IDs or multiple dates, but this is not the case for this dataset.

```python
# Identify outliers using IQR

numeric_cols=df.select_dtypes(include=['number']).columns.tolist()

print("\nIdentifying outliers using IQR:")
outlier_indices = {}
for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)].index
    outlier_indices[col] = outliers
    print(f"  {col:>20}: {len(outliers)} outliers identified")

# Combine indices from all columns
all_outlier_indices = [idx for indices in outlier_indices.values() for idx in indices]
all_outlier_indices = list(set(all_outlier_indices)) # Remove duplicates

print(f"\nTotal unique outliers identified across numeric columns: {len(all_outlier_indices)}")
```

```
Identifying outliers using IQR:
                  id: 0 outliers identified
               price: 1146 outliers identified
            bedrooms: 546 outliers identified
           bathrooms: 571 outliers identified
         sqft_living: 572 outliers identified
            sqft_lot: 2425 outliers identified
              floors: 0 outliers identified
          waterfront: 163 outliers identified
                view: 2124 outliers identified
           condition: 30 outliers identified
               grade: 1911 outliers identified
          sqft_above: 611 outliers identified
       sqft_basement: 496 outliers identified
            yr_built: 0 outliers identified
        yr_renovated: 914 outliers identified
             zipcode: 0 outliers identified
                 lat: 2 outliers identified
                long: 256 outliers identified
       sqft_living15: 544 outliers identified
          sqft_lot15: 2194 outliers identified

Total unique outliers identified across numeric columns: 6679
```

7.4 Identifying Outliers and Removal:

In this analysis, there were 6,679 different outliers across the quantitative-related variables. The variable with the highest contributions were sqft_lot (2,425) sqft_lot15 (2,194), view (2,124), and price (1,146). These outliers are representative of a small subset of properies, which have very large lot sizes, great views, or high prices.. The variables floors, zipcode, and yr_built do not have any outliers, indicating that these

distributions are steady across all average properties. Moderately extreme outliers in sqft_living, grade, and bathrooms could describe luxury properties or properties with increased square footage.

```python
# Cleaning the Outliers (Masking Renovation Year and Waterfront)

protect_mask = (df['yr_renovated'] > 0) | (df['waterfront'] > 0)

# Drop only rows that are outliers AND NOT protected
to_drop_mask = df.index.isin(all_outlier_indices) & (~protect_mask)
df_clean = df.loc[~to_drop_mask].reset_index(drop=True)

```

```python
df_clean.shape
```

```
(15969, 21)
```

Because waterfront and yr_renovated merit high value and are infrequent in that value range, simply removing all of their outliers would be equivalent to removing those premium listings from the data altogether. That is precisely why they are masked. Masking logically protects those rare high-end properties that legitimately drive higher price thresholds. It also serves a business purpose and maintains representation in the luxury or renovated segments of the market, which are fundamental to price prediction and policy decisions.

8. Feature Engineering :

The purpose of this code is to provide a standard feature engineering pipeline for creating new

```python
def feature_engineering(df_clean):
    df_fe = df_clean.copy()

    if 'date' in df_fe.columns:
        df_fe['sale_date'] = pd.to_datetime(df_fe['date'], errors='coerce')
        sale_year = df_fe['sale_date'].dt.year.fillna(2025)
    else:
        sale_year = pd.Series(2025, index=df_fe.index)

    df_fe['House_Age'] = sale_year - df_fe['yr_built']
    df_fe['Living_to_Lot_Ratio'] = df_fe['sqft_living'] / df_fe['sqft_lot'].replace(0, np.nan)
    df_fe['Basement_Present'] = (df_fe['sqft_basement'] > 0).astype(int)
    df_fe['Baths_per_Bedroom'] = np.where(df_fe['bedrooms'] > 0, df_fe['bathrooms'] / df_fe['bedrooms'], 0.0)
    df_fe['Renovated'] = (df_fe['yr_renovated'] > 0).astype(int)
    df_fe['Years_Since_Renovation'] = np.where(df_fe['yr_renovated'] > 0, sale_year - df_fe['yr_renovated'], 0)

# Basic hygiene
    df_fe.replace([np.inf, -np.inf], np.nan, inplace=True)

    return df_fe


# Apply feature engineering
df_fe = feature_engineering(df_clean)

```

variables that have economic meaning out of the cleaned data set. Each engineered variable demonstrates an important real estate or pricing quality.

Several useful variables are provided:

- **sale_year:** We converted date to a proper datetime to get the sale year. In any cases where date was missing, we gracefully filled with a neutral proxy (2025) to allow formulae to calculate successfully. The sale_year variable was then used to derive House_Age = sale_year - yr_built and Years_Since_Renovation = sale_year - yr_renovated (else 0).
- **House_Age** is important to determine how old a property is, and newer properties will keep more value and depreciate less.

- **Living_to_Lot_Ratio** conveys land-use efficiency, and reveals whether the property is fulfilling maximum living area relative to its lot size; which is related to urban premium areas.
- **Basement_Present** (1 = yes, 0 = no) reveals whether the property has a basement, as additional livable space will typically be reflected by higher price and comfort.
- **Baths_per_Bedroom** determines the quality and balance of home layout and comfort, showing whether the property is representing a higher-end design.
- **Renovated** shows upgraded homes compared to older homes.
- **Years_Since_Renovation** indicates how long ago the renovation occurred; higher improvements to properties tend to have stronger price implications when they were more recently renovated.

In addition to the new feature implementation, we had also performed data cleansing in the function, where we replaced infinite, or missing, computed values with NaN to avoid instability in the model output.

9. Exploratory Data Analysis (EDA) on Clean Dataset:

```python
import seaborn as sns
import matplotlib.pyplot as plt

# Select numeric columns for correlation
numeric_corr = df_fe.select_dtypes(include=['int64', 'float64'])

# Compute correlation with price
price_corr = numeric_corr.corr()['price'].sort_values(ascending=False)

# Display correlations
print("Correlation of features with price:")
print(price_corr)

```

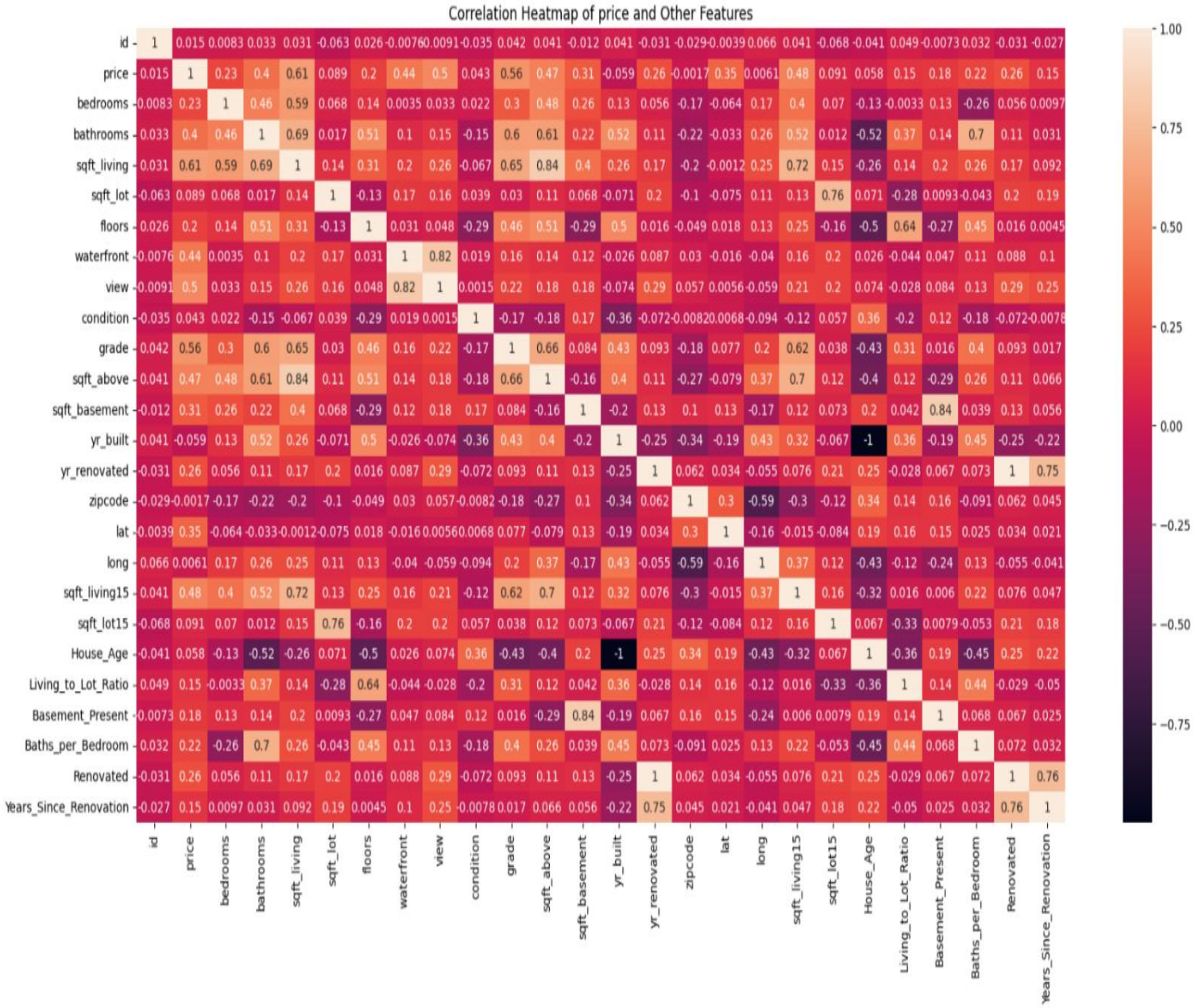
```
Correlation of features with price:
price                     1.000000
sqft_living               0.607239
grade                     0.558069
view                      0.502386
sqft_living15             0.478034
sqft_above                0.470006
waterfront                0.437224
bathrooms                 0.396043
lat                       0.346003
sqft_basement             0.309605
yr_renovated              0.261399
Renovated                 0.260872
bedrooms                  0.231181
Baths_per_Bedroom         0.220111
floors                    0.195794
Basement_Present          0.178746
Years_Since_Renovation    0.153208
Living_to_Lot_Ratio       0.149684
sqft_lot15                0.091481
sqft_lot                  0.088686
House_Age                 0.058471
condition                 0.043094
id                        0.015075
long                      0.006109
zipcode                  -0.001685
yr_built                 -0.058537
Name: price, dtype: float64
```

9.1 Correlation Matrix and Heatmap:

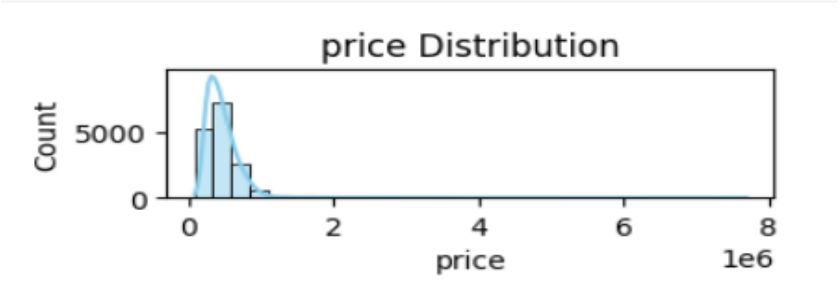
The correlation display table and correlation heatmap show some relevant characteristics. The living area (sqft_living) is the strongest predictor of price (r ≈ 0.61), which is not surprising, and larger homes are worth more. The next strongest predictors correlate with the property's grade (0.56) and whether it has a view (0.50). Living area and its associations highlight how the home's construction quality and the view tremendously contribute to property pricing . The sqft_above, sqft_living15, and waterfront features maintain a moderate degree of correlation (0.43–0.47), which takes spatial and location considerations into account. The newly engineered features clearly added some increased explanatory value, given the positive correlations with yr_renovated and Renovated both (~0.26) and the feature performance would suggest that more modern homes tend to be worth more. The Baths_per_Bedroom (0.22) and Living_to_Lot_Ratio (0.15) show a weaker, but still reasonable

feature contribution correlation behavior of the home valuation. These would suggest some design to be more efficient or utilizing land area more extensively must also modestly correlate to more value or increased pricing. Interestingly, the House_Age feature displays a mild negative correlation (–0.06), which suggests newer properties tend to be worth more, but again the increase in value is only mildly correlated with consideration of renovation.

```
plt.figure(figsize=(20,10))
sns.heatmap(numeric_corr.corr(), annot=True)
plt.title("Correlation Heatmap of price and Other Features")
plt.show()
```



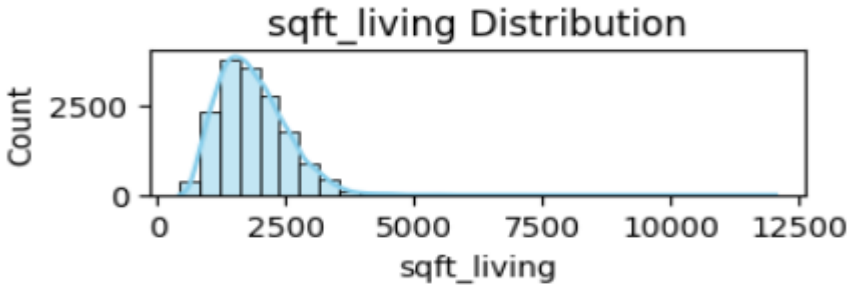
9.2 Histogram of Key Variables:



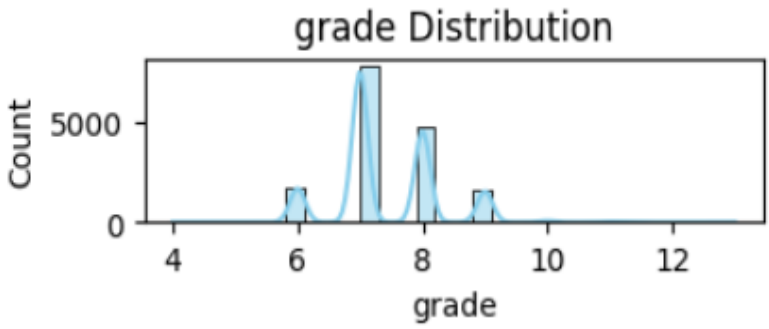
- **Price:**

The price distribution has a right skew, indicating that many homes are priced less than 1 million USD while there are a few luxury high-end listings that can be priced at 7-8 million. This right skew shows there is a very limited number of homes priced in this high-end luxury segment that are influencing the mean price.

- **Sqft_Living:**



Sqft_living also follows a right skew, suggesting that there are many homes that are average in size, but there are also a few very large homes that are pulling the upper tail out. Since there is a high correlation between sq-ft-living and price, this is one of the more important features influencing housing value.

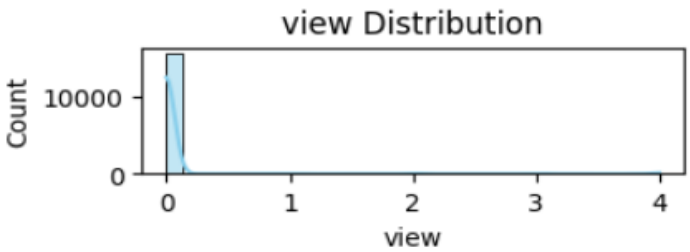


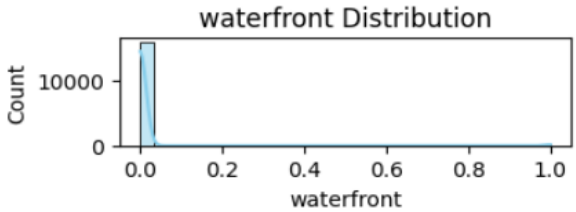
- **Grade:**

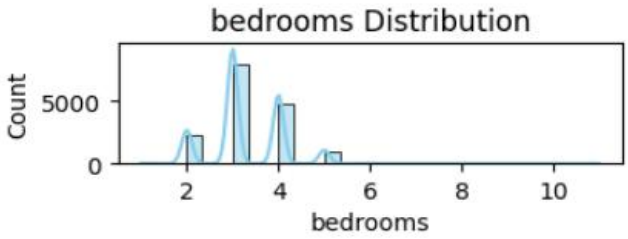
The grade variable has a nearly normal distribution ranging in the mean from 7 to 8. This suggests that most homes have average to above-average quality. Many of these higher highly ranked homes will heavily influence a price premium.

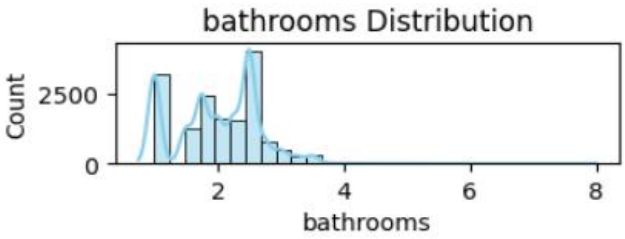
- **View and Waterfront** :

Both variables have are heavily left skewed indicating that most homes do not have premium views or waterfront homes. These are extremely rare features, but very valuable. This is likely why they may have remained during outlier removal.



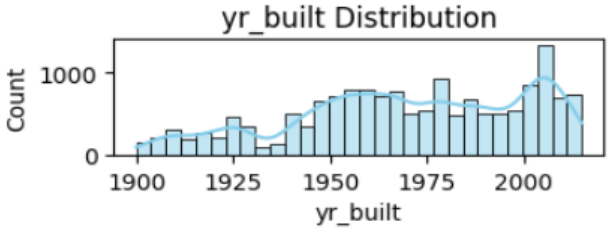


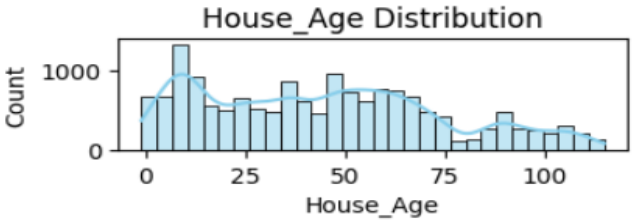




- **Bathrooms and Bedrooms** :

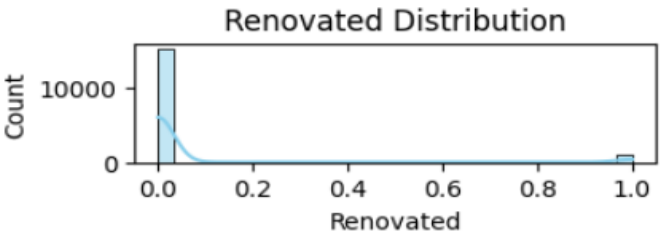
Bathrooms and bedrooms exhibit slight right skewness, clustering around 2–3 bathrooms and 3–4 bedrooms. This suggests a market for standard single-family homes with limited multi-family or high-end properties at the upper end.

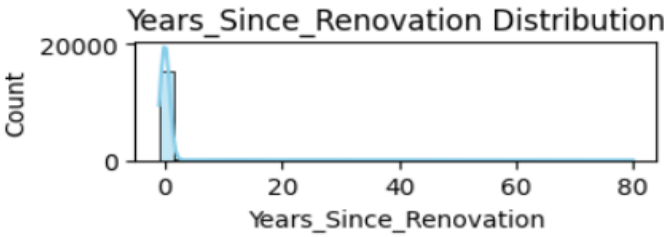




- **Yr_Built and House_Age:**

Yr_built increases sharply after 1950, peaking around 2000 — indicative of a modern housing boom — while House_Age demonstrates wide variability and perhaps moderates the market, supporting both the old and new housing stock. This variation allows us to model depreciation and renovations.

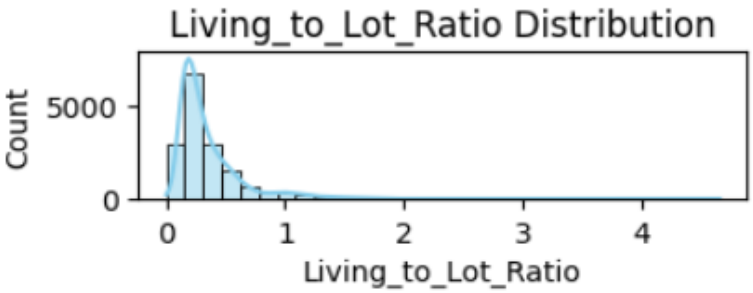




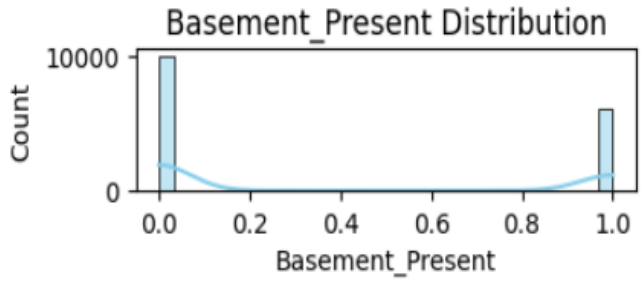
- **Yr_Renovated and Years_Since_Renovation:**

Both variables exhibit extreme skew to zero indicating that most homes have never been renovated, while extremely renovated homes represent a viable high-value segment — as prices typically respond well to recent renovations.

- **Living_to_Lot_Ratio:**



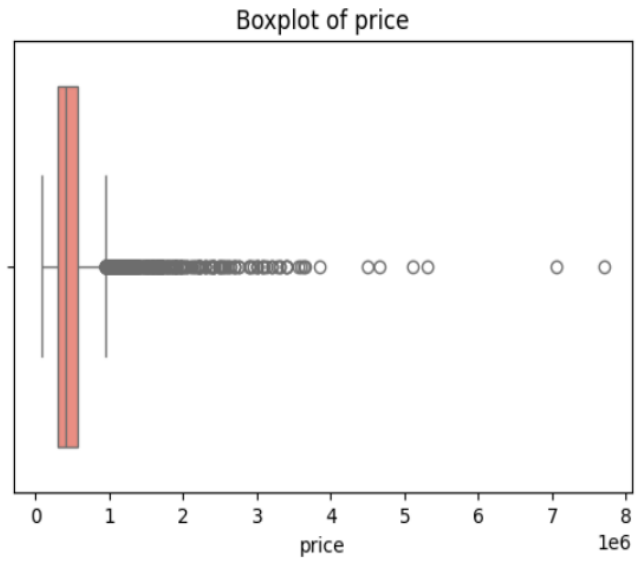
This engineered variable also exhibited right skewness. Higher ratios indicate more intense land use, which is typical in urban or high demand regions where smaller lots have larger living space. This types of property tend to The homes with a higher price per square foot..

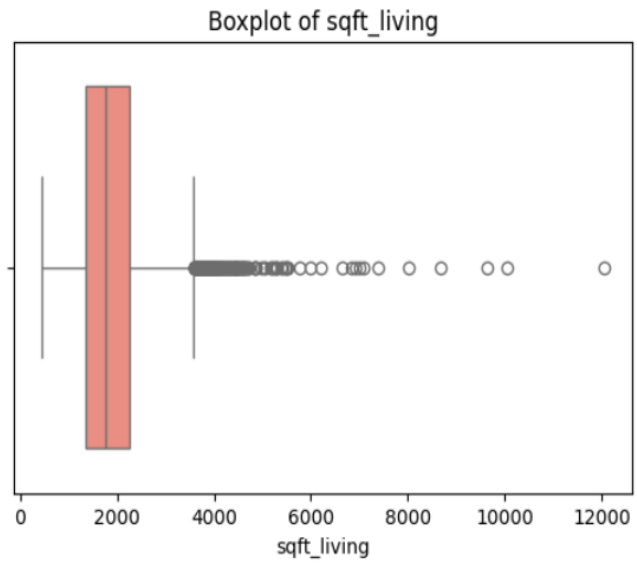


- **Basement_Present:**

The distribution indicates that a smaller percentage of homes in the area had basements. While not every home in the area has a basement, having one creates additional usable space and generally creates comfort. Basements are associated with higher-end urban residences or in more extreme architecture meant for very cold climates.

```
for col in ['price','sqft_living','bedrooms','bathrooms']:
    plt.figure(figsize=(6,4))
    sns.boxplot(x=numeric_corr[col], color='salmon')
    plt.title(f"Boxplot of {col}")
    plt.show()
```





9.3 Boxpot of Key Variables:

- **Price:**

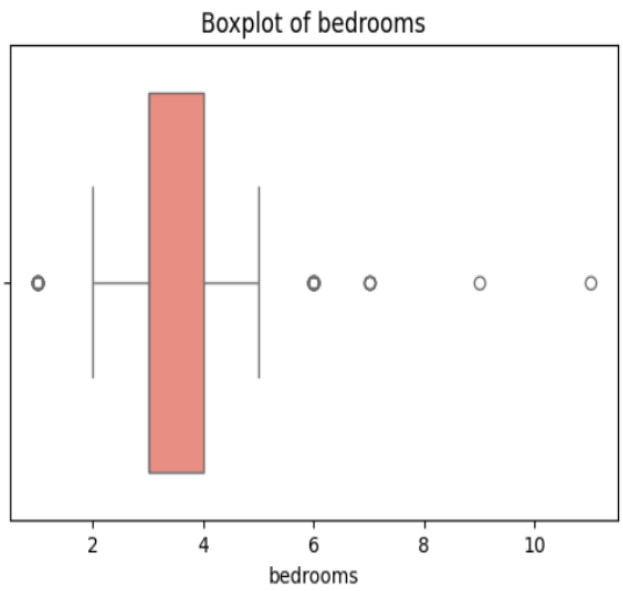
Similar to the Histogram the boxplot of price distribution indicates a very strong right skew with a large mass of homes below 1 million dollars and a long tail of outliers growing from 7–8 million. These luxury or waterfront properties are rare, but when total outliers are still taken in consideration with the mass, they disproportionately skew the effective price distribution based on these outliersderiving a mean price reflective of those luxury properties or advanced waterfront properties.

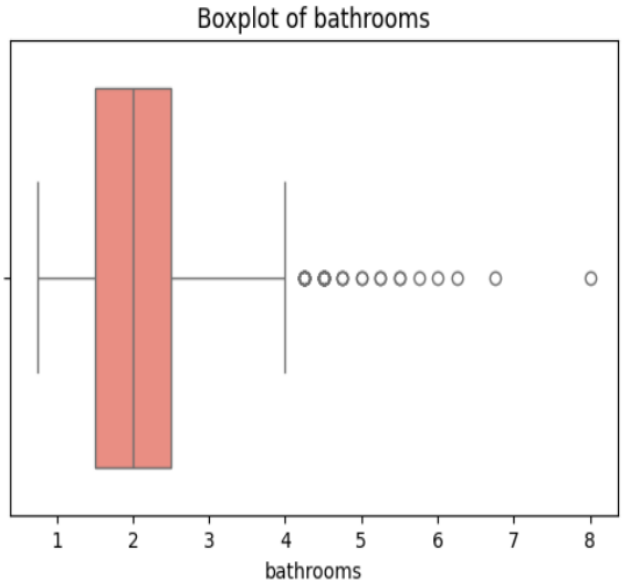
- **Sqft_Living**

As with the price feature, sqft_living is also right skewed, which we can relate to, with most homes being more modest and some larger homes (bigger than 5,000 sq. ft.) are outliers. Since this feature had the highest degree of correlation with price, we shall keep these outliers (after we have already removed the extreme anomalies), in order to ensure that the model effectively reflects size of the property impacting the valuation of the property.

- **Bedrooms**

The boxplot displaying bedrooms depicts a modest spread, most commonly between 2 and 4 bedrooms, with a few embodiments consisting of 8-10 bedrooms that seem to remark upon multi-family and luxury properties. Although infrequent, they are legitimate observations and should not be excluded from analyses, as the decision to include or exclude those values can shift analyses.



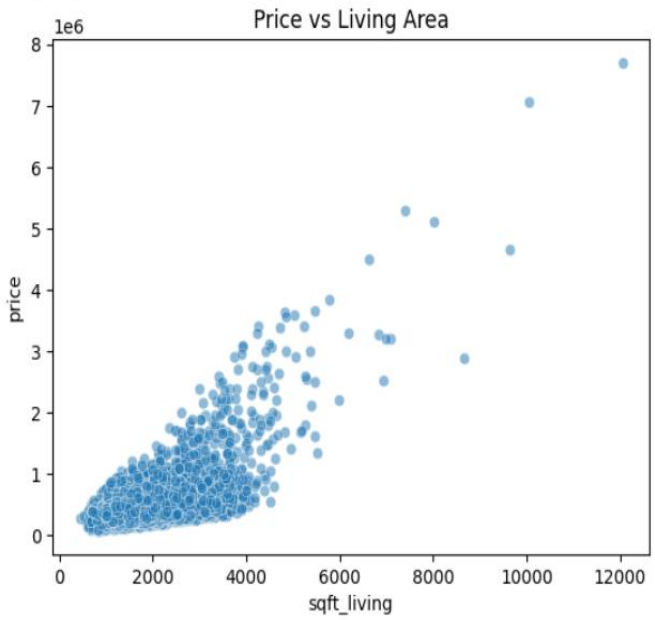


- **Bathrooms**

The same general shape appears in the bathrooms, with a relatively symmetric distribution around 2-3 bathrooms. We see a few high-end outliers at 6-8 bathrooms. This variable is valuable in terms of predicting comfort and design premium and often acts as a market price variable influencing the mid- to upper-tiers of the market.

```
sns.scatterplot(x='sqft_living', y='price', data=numeric_corr, alpha=0.5)
plt.title('Price vs Living Area')
```

```
Text(0.5, 1.0, 'Price vs Living Area')
```



9.4 Bivariate Analysis:

9.4.1 Scatterplot Price vs Living Area:

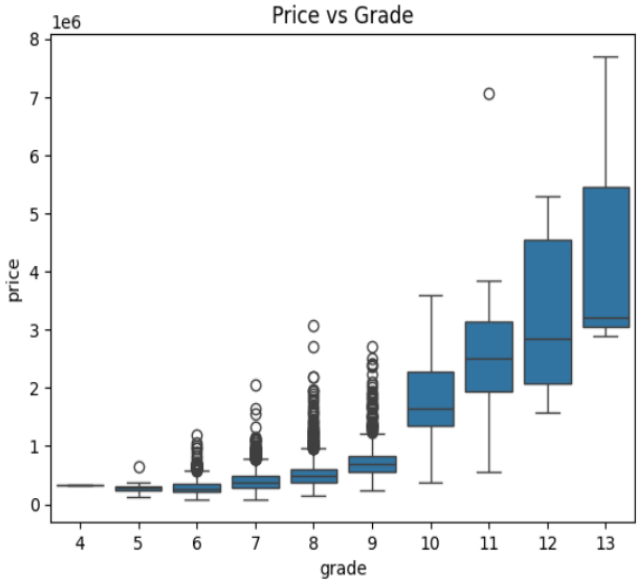
The scatterplot indicates a strong positive association–as living area rises, property price also rises. Most of the homes tend to cluster beneath 3,000 square feet and below $1 million, but there is still considerable upward vertical spread for the larger homes, even up to $7-8million. The relationship appears to be non-linear in nature, suggesting that price growth increases at much higher rates for larger homes and that space can contribute to useful increased value in the upper end of the housing

market. Additionally, there are still very few high-price scattered observations, which likely indicates their worth as luxury properties for example or waterfront properties. Both do command higher prices that exceed what growth rates warrant on any basis of sheer space alone.

9.4.2 Boxplot Price vs Grade:

```
sns.boxplot(x='grade', y='price', data=numeric_corr)
plt.title('Price vs Grade')
```

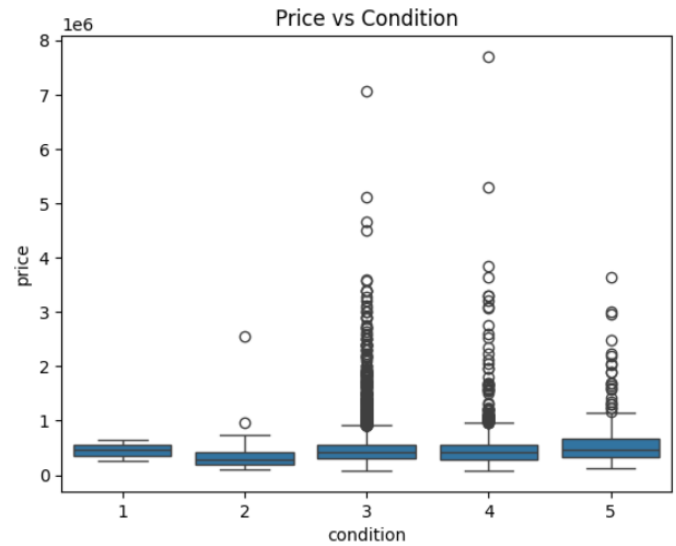
```
Text(0.5, 1.0, 'Price vs Grade')
```



The boxplot effectively displays a strong positive association between grade and price. Homes are grouped into lower grades (4–7), with associated price ranges showing relatively low prices and tightly clustered prices, followed by gradually increasing prices and more variable range as the grade improves. From grade 9, where the boxplot begins to show a median price shift upward, median prices for grades 11–13 show the greatest range of price, often exceeding USD 3–4 million. This is consistent with higher construction quality, premium materials, and customized architectural design, each significantly influencing market value. The presence of a few high-priced outliers in each of the lower grade ranges applies to local demand differences or additional luxury features were added.

```
sns.boxplot(x='condition', y='price', data=numeric_corr)
plt.title('Price vs Condition')
```

```
Text(0.5, 1.0, 'Price vs Condition')
```



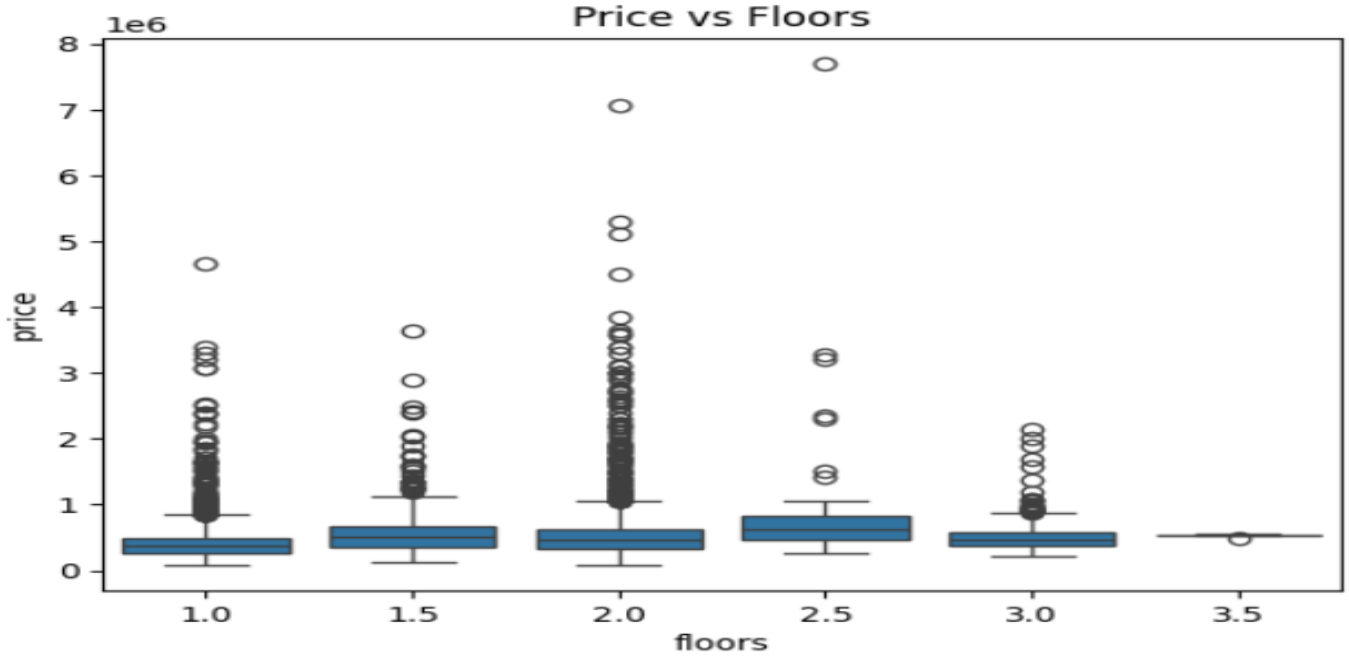
9.4.3 Boxplot Price vs Condition of the House:

The boxplot depicts a clear upward inclination; higher-grade homes exhibit systematically higher price statistics. Low-grade homes (4-7) display narrow price ranges, indicating more uniform, and lower value properties. The median and range of homes' price increase substantially as grade increases, grades 11-13 will include a luxury portion as prices exceed USD 3-4 million. This pattern showcases a positive correlation between the quality of homes and market value. It also implies that upper grades contain greater variability of prices suggesting that luxury homes are likely to vary in location, design, colour, or amenities.

9.4.4 Boxplot Price vs Floors in the House:

```
sns.boxplot(x='floors', y='price', data=numeric_corr)
plt.title('Price vs Floors')
```

```
Text(0.5, 1.0, 'Price vs Floors')
```

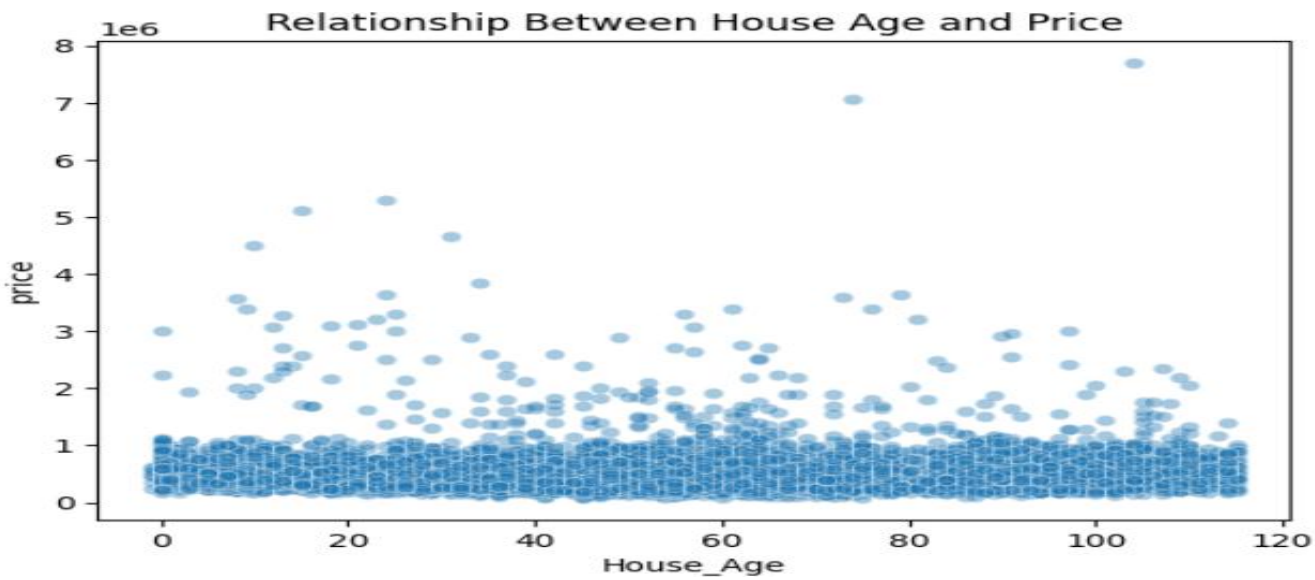


The chart indicates that there is no substantial price increase with additional floors. Homes, whether they are 1-, 2-, or 3- floored, are mostly clustered around the same median price — generally below 1 million USD. There are one or two extreme cases at 2- and 2.5-floored homes, indicating expensive or custom multi-level homes, but are atypical at point of sale. This indicates that price is likely not driven solely by the number of floors — price is driven more by other amenities/features associated with the home, such as living area, grade, views or location.

9.4.5 Scatterplot of House Age and Price:

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.scatterplot(x='House_Age', y='price', data=numeric_corr, alpha=0.4)
plt.title('Relationship Between House Age and Price')
plt.show()
```

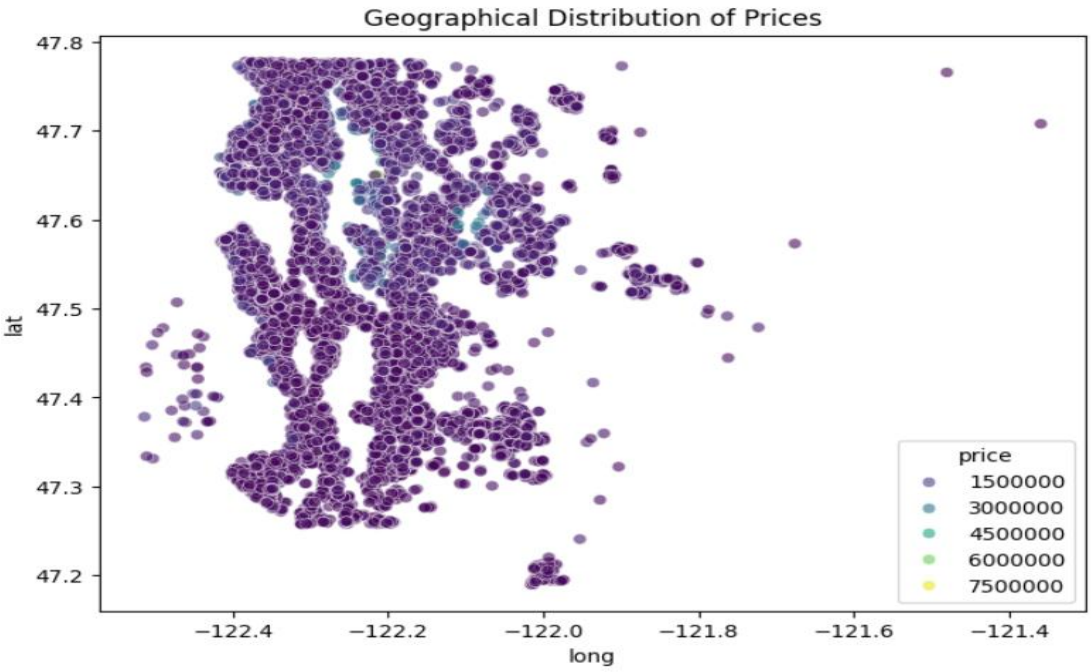


The scatterplot shows a negative trend between the age of a house and price of the homes. Most

homes will be below 1 million USD regardless of the age, but younger homes are more likely to be toward the higher price side. This indicates that while the home is still functioning properly, newer homes may sell at higher prices due to newer designs and materials, and less maintenance work. However, even in the distributions of the older homes where the density is lower, there are older homes that sold at higher prices, which can be attributed to improvements, location, or historical significance. There is some weak correlation overall, which implies that age is not the only determinant of price. While age is correlated to price, there are surely interactions between age and other features that also affect price, such as renovations, location, and condition.

9.4.6 Geographical Distribution of Prices:

```
plt.figure(figsize=(8,6))
sns.scatterplot(x='long', y='lat', hue='price', data=numeric_corr, palette='viridis', alpha=0.6)
plt.title("Geographical Distribution of Prices")
plt.show()
```



The scatterplot shows a distinct spatial clustering of prices. For example, more expensive homes (lighter colours such as green and yellow) often cluster within definable geographic areas, while less expensive homes (dark purple) fill the more general area. High-value clustering is probably associated with prime coastal or central city locations, where demand is highest and land is limited. As the coordinates track easterly (increasing longitude) or southerly (decreasing latitude), the density of high-priced homes decreases, suggesting that prices fall farther away from the core or waterfront. This indicates proximity is a major factor in am housing prices, relative to facilities, urban centers, or attractive ‘views’.

9.4.7 Scatterplot of Price vs Age of House Coloured by Renovation:

```
sns.scatterplot(x='House_Age', y='price', hue='Renovated', data=numeric_corr, alpha=0.5)
plt.title('Price vs Age (Renovated vs Non-Renovated)')
plt.show()
```

The data indicates that renovated homes (orange points) tend to be priced higher than non-renovated homes (blue points), independent of age. Both old and new properties have higher prices when they are renovated, and consequently, the typical decline in prices associated with older homes is reduced. Non-renovated homes tend to group closely together below USD 1 million, but renovated homes are more broadly represented, extending much beyond USD 2–3 million, and even USD 5 million for a few luxury properties. Notably, older properties (age greater than 50 years) that are renovated maintain their value or are valued greater, generally affirming that quality renovations can restore value to older housing assets.

9.5 Applicability for Business:

It is essential for real estate stakeholders (developers, investors, brokers, and lending institutions) to understand what affects housing prices. The visualisations demonstrate how the features of the housing stock, including square footage, quality of construction, age, renovation status and location have a differential effect on value.

From a business point-of-view:

Increasing revenue: Developers and investors can look at the attributes that are associated with a price premium (e.g., square footage, quality level, waterfront setting, recent renovations) which will allow them to focus investment in segments that return a higher yield.

Cost savings: By understanding which features do not affect value (e.g., number of floors alone), companies may decide to avoid spending on features or renovations that the market does not reward, saving themselves unnecessary renovations or construction costs.

Precision pricing: This also allows real estate agents to arrive at more precison listing prices based on a better understanding of the influences—attributes of size, quality and location—allowing them to better reach and negotiate an acceptable price for listings and improve sales and sell outs while shortening the time to sell the property.

Value experience agitation: The buyer, or renter, gains a measure of transparency in the process as they can know which attributes are important and factor that into their decision on which value to select, as well as be not be worried about which valuation process was used.

Mitigation of risk: Lenders and insurers can better assess residential property risk and ascertain the value of collateral by understanding the attributes impacting the value of collateral property, given possible over-valuation of an asset where lenders are exposed.

10. Implementation and Results:

All models were implemented in Python using libraries like scikit-learn, pandas, and matplotlib. The dataset was first cleaned and enhanced through feature engineering. New variables such as *house age, living-to-lot ratio, years since renovation,* and *basement presence* were added to capture structural and quality differences between homes. These engineered features help the model reflect what actually drives price, things buyers and investors care about, like how new a home feels or how efficiently the land is used.

Once the data was ready, two sets of models were built, one for price prediction (regression) and another for luxury classification. The regression models aimed to estimate the continuous price value, while the classification models were trained to predict whether a property fell above or below the “luxury” threshold of $560,000, which represented roughly the top 25% of the market. This setup addressed two business needs at once: accurate pricing for everyday listings and quick identification of high-end homes for focused marketing or investment decisions.

10.1 Model Implementation:

The following algorithms were used for the regression task: Linear Regression, Random Forest, and Gradient Boosting Regressor while for the classification task, the following algorithms were trained on the data: Logistic Regression, Random Forest Classifier, and Gradient Boosting Classifier.

All modelling proceeded by fitting on 80% of the data and testing on the remaining 20% of data. Data transformations e.g. pipeline was created to standardize numerical features and apply scaling. Tree-based models like Random Forest and Gradient Boosting used raw numeric inputs since they handle feature scales internally, while Linear and Logistic Regression models included a standard scaler.

10.2 Regression Results:

The models were evaluated using three main metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared. MAE and RMSE indicate how far predicted prices were from actual prices, and R-squared tells us how much of the variation in price was explained by the model. Lower error values and higher R^2 indicate better performance. From a business point of view, a lower MAE means agents and sellers can set prices that are both competitive and realistic. It helps avoid underpricing or overpricing homes and keeps sales cycles shorter.

```
Regression comparison:
                          MAE           RMSE        R2
RandomForest      102072.691419  145617.530877  0.662846
GradientBoosting  107154.705346  149723.552053  0.643565
LinearRegression  120060.238049  173799.946242  0.519714

Best Regressor: RandomForest
```

Table 1: Regression Model Performance Comparison

Out of the three models, **Random Forest** performed the best. It had the lowest prediction errors and the highest R^2 score, explaining about 66% of the price variation. The findings demonstrate that Random Forest understands the combination of house attributes and costs more than Linear Regression, which has a failure to capture complicated patterns. This kind of precision gives agents a good baseline when determining list prices, and gives investors' confidence that they can determine whether a house is over or under valued.

10.3 Classification Results:

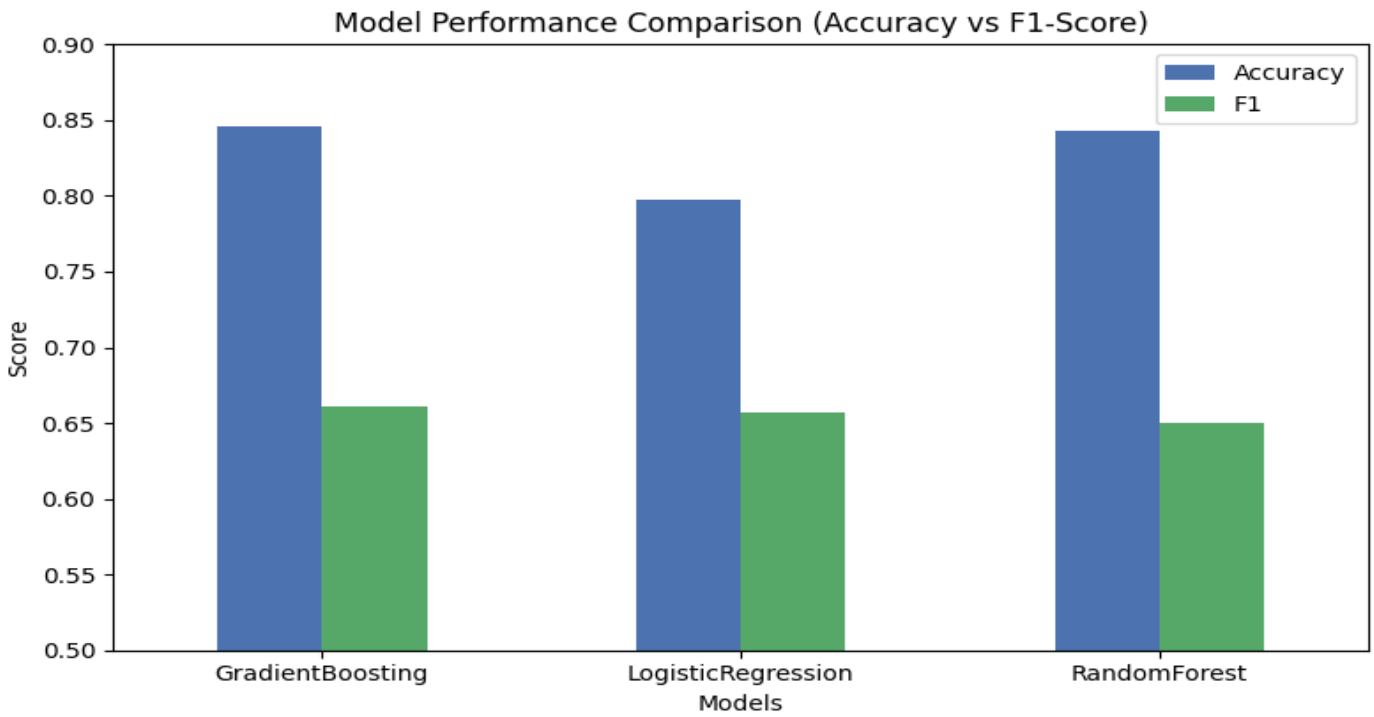
```
Luxury threshold: $560,000

Classification comparison:
                     Accuracy  Precision    Recall        F1   ROC_AUC
GradientBoosting     0.845648   0.728788  0.605031  0.661168  0.889169
LogisticRegression   0.797746   0.568286  0.779874  0.657476  0.883933
RandomForest         0.842830   0.729264  0.586164  0.649930  0.890334

Best Classifier: GradientBoosting
```

Table 2: Classification Model Performance Comparison

We evaluated all classification models on five different metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC. Accuracy tells us the overall percentage of correct predictions. Precision indicates how many of the homes predicted as luxury were indeed luxury. Recall tells us how many actual luxury homes the model correctly predicted. F1-score combines Precision and Recall into a single, well-balanced measure and ROC-AUC indicates how well the model classifies luxury from non-luxury homes. The higher these values are the better performance of the model.



Comparison of Model Performance (Accuracy vs F1-score)

The figure compares the performance of every classification model regarding predicting luxury homes. The Gradient Boosting Classifier was the model that performed the best overall, and it offered the best accuracy and best F1 score (balance between Precision and Recall). So, you could expect that model to identify the homes that are priced to be in the highest tier. That is useful

information for real estate teams to direct their advertising and resources to the houses with the highest return on investment potential. The Gradient Boosting Classifier provided the most reliable results through a high accuracy and strong balance of Precision and Recall. Random Forest performed a close second, and Logistic Regression performed the weakest among the classifications as it is a simpler model and can't capture the non-linear relationships seen in the luxury housing market.

10.4 Interpretation of Results:

The takeaways from the regression and classification models suggest a common theme: tree-based ensemble approaches outperform simple linear models in a consistent manner. In practical ways, tree-based methods can identify more complex, non-linear relationships among the variables such as size, bedrooms, grade, condition, and location. Each variable rarely affects price in isolation in a real estate context, which is likely why Random Forest and Gradient Boosting models report stronger outcomes. This reinforces how home values are the intersection of multiple variables rather than a singular characteristic. It also aligns well with what real homebuyers consider, such as quality, location, and design, as a collective unit.

The regression findings indicate that Random Forest yielded the least prediction error and explained approximately 66% of variation in house prices. This demonstrates an ability to predict price movement at a high level of accuracy which serves as a valuable tool for property valuation and investment analysis. Gradient Boosting Regressor also performed well but was notably less accurate than Random Forest, while Linear Regression did not accurately model the complexity often seen in the real world of property pricing.

The results of the classification task showed a clear winner in the Gradient Boosting Classifier results, with approximately 84.6% accuracy and a ROC-AUC of 0.889. This indicates it is effectively distinguishing between luxury housing stock and regular housing stock. Random Forest follows closely behind with Logistic Regression coming in third, missing more true luxury cases. Overall, Random Forest is greatest for predicting continuous housing prices while Gradient Boosting performs the best for identifying luxury housing. Together, these two models represent a full solution set for the real estate business. One calculates fair market prices while the other flags luxury homes, enabling teams to make more informed pricing, investment, and marketing decisions.

10.5 Gradio:

This Gradio application creates an interactive interface with a real estate machine learning model which allows users to evaluate house properties about house prices and classifications. The system

is built by taking specific housing features such as house size, number of rooms, house age etc. and processing them with two trained models.

The predict price function takes features like house age, size, and number of bedrooms, then uses a pre-trained regression model to predict the estimated price. Plus, some adjustment for Gradio is made for a more realistic result based on real life scenarios, such as decreasing the value for older homes and adding a premium price for the recently renovated houses, which made the predictions more realistic rather than just generate some fixed modelling results.

The second function is using classification which helps to determine whether a house is classified as a “Luxury” property. It takes the same features from the regression model after adjustments on the house age and renovation status to calculate a probability score. Users can distinguish if the house is luxury with a simple output whether the house is luxury or not and a confidence score behind it.

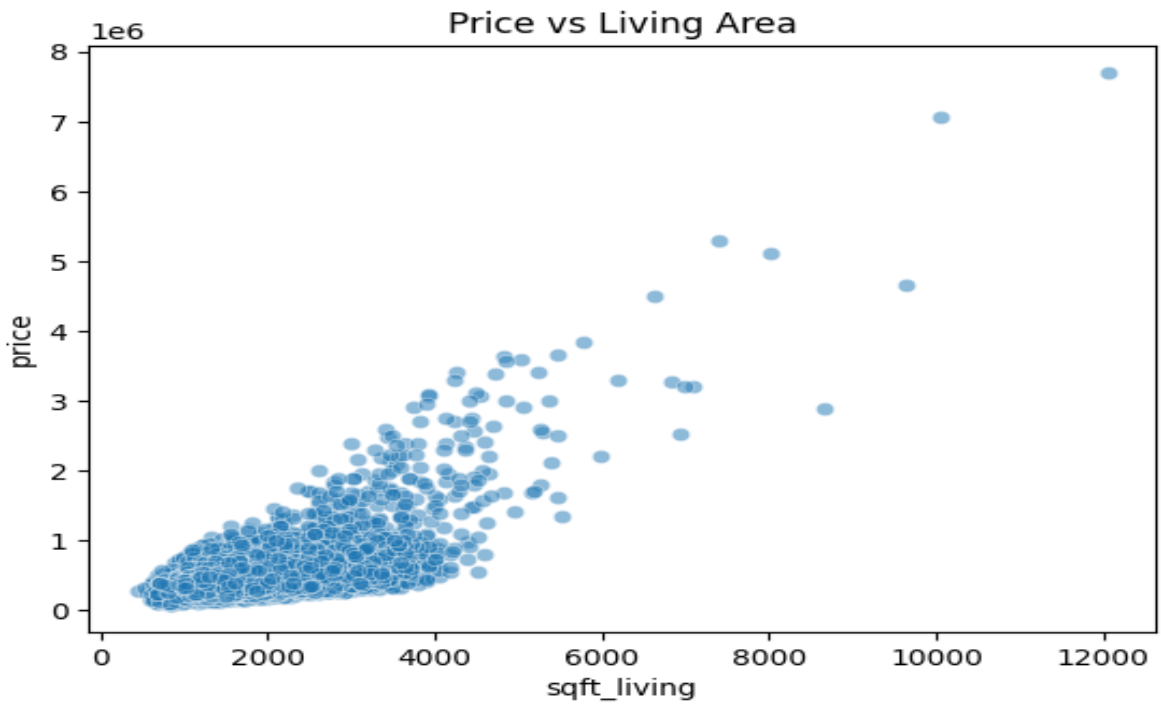
The Gradio interface is presented and organized into two tabs. The price prediction tab lets users input details to get a predicted price while the luxury classifier tab can give the market segment analysis about the stats of houses. By simply choosing the suitable features, results are generated immediately and help users easily get data-driven insights without any difficult operation needed. This demonstrates the power of machine learning to transform complex data into practical tools for decision-making in real estate.

11. Business Insights and Recommendations:

The models' results offer the clearest insight into the drivers of housing prices in King County as well as how these insights can help agents, investors, and homeowners make more informed decisions. The goal was simple: test whether machine learning could predict fair property prices and identify luxury homes at more reliable rates than traditional methods of valuation.

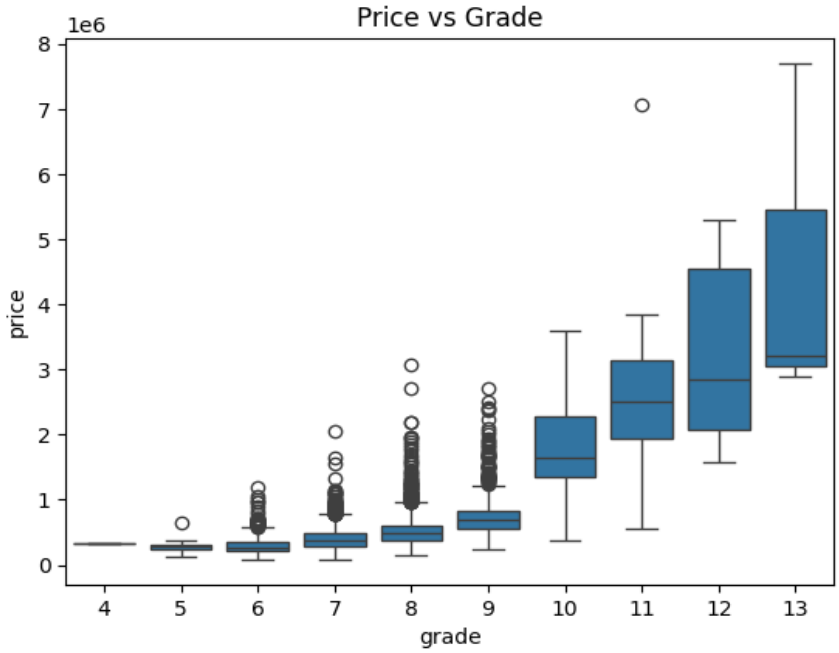
11.1 Key factors influencing price:

The data made it obvious that some features carry far more weight than others. The factors that were revealed to be the most important price indicators were living area, construction quality, condition, renovation status, and location. Generally speaking, larger quality houses sell for more money, while older or poorly maintained houses lose value. Houses that have been renovated also typically have better resale value, particularly if the renovations are relatively new.



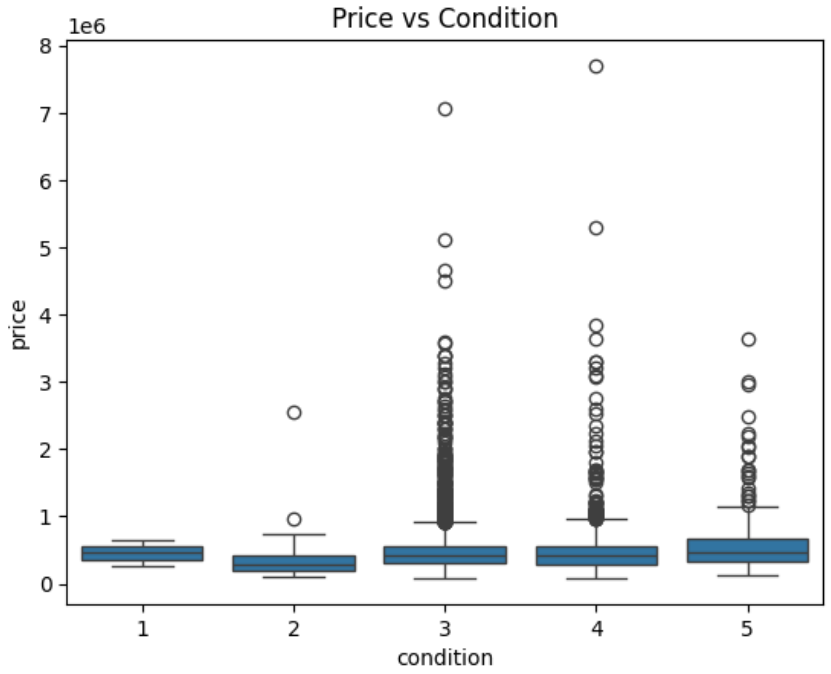
Price vs Living Area

As living area increases, price climbs too. The trend is strong, though prices start to flatten for very large homes. This tells sellers that expanding or improving living space can raise property value, but only up to a point (around 4000 square feet and above).



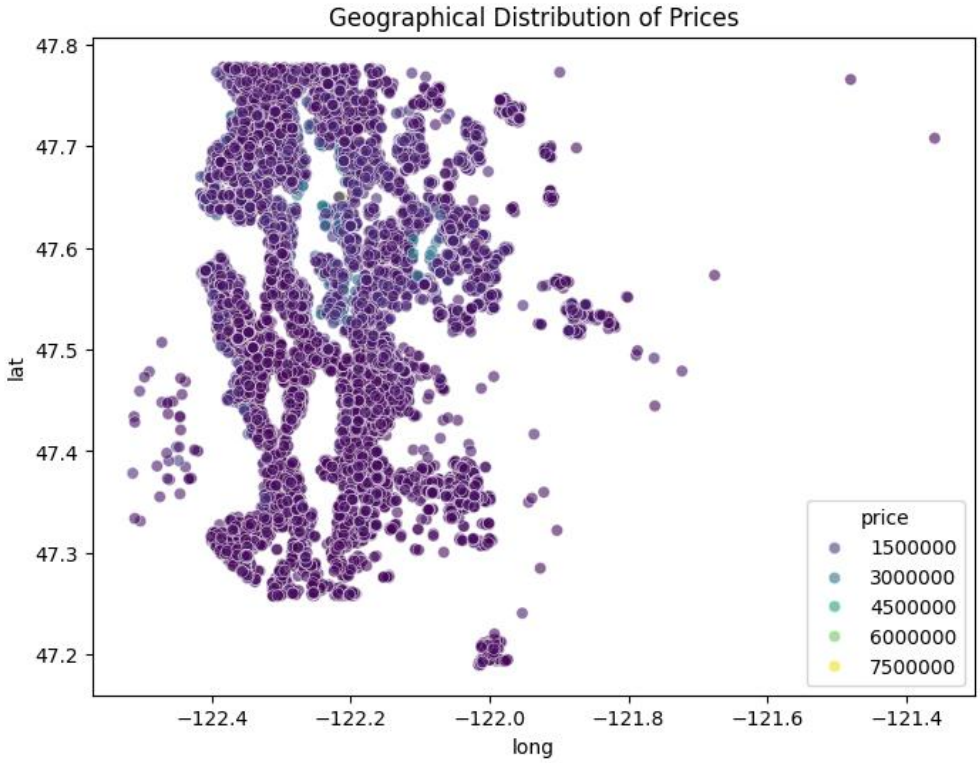
Price vs Grade

Noticeable jumps in median price occur with each construction grade purchase. Higher quality materials and finishes are worth the cost to buyers, showing that overall quality is just as important as size.



Price vs Condition

Homes that are in better condition, accompany a higher price, but have less impact than grade or location. This is an indication that while condition is important, the quality and structural design appears to outweigh it.



Geographical Price Distribution

Price clusters appear in specific areas, mostly near central or coastal locations. It confirms what agents already know: location drives price more than almost any other feature.

12. Business Applications and Strategic Value:

The findings from both the models and the plots can be directly applied across the real estate market. The patterns seen in the visuals, such as the steady rise in price with larger living areas and higher grades, are the same relationships that allowed the models to make accurate predictions. This connection between data exploration and model performance gives these insights real practical value.

For **real estate agents**, the models offer a data-driven way to set realistic listing prices. Instead of relying only on experience or comparable sales, they can use the predicted prices as a reference. The “Price vs Living Area” and “Price vs Grade” plots help agents explain to clients why two houses that appear similar can still have very different market values. This transparency helps build trust and reduces pricing disputes.

Investors can identify properties that are underpriced for their size, quality, or location through the scatterplots and models outputs. Properties with these characteristics can often be renovation opportunities where selective improvements can be made to maximize resale value. Not only does it streamline the decision-making process on where to invest, but also makes it more valuable economically.

Additionally, **Sellers** can also use the same data to identify which upgrades will provide the most value. The “Price vs Condition” and renovation-related findings show that buyers care more about lasting quality than small decorative changes. Upgrading structure, finishing, or functionality is more likely to raise property value than quick cosmetic fixes. While **buyers** can use the predicted prices from the models as a checklist against listing prices. Comparing the two helps them avoid overpaying and gives them a stronger position when negotiating. It also allows them to focus on properties that are fairly priced for their size, grade, and location.

13. Limitations and Future Improvements:

While working on the models, a few limitations became clear. The dataset only covers King County, so it feels very specific. The models learned the local market patterns well, but the same relationships might not be held in other areas with different price ranges or living standards. It also became obvious that some important details were missing, such as school ratings, proximity to downtown, or neighbourhood safety. These are the kinds of variables we start wishing we had once we saw how much the model depends on what is available.

Each model brought its own challenges. Random Forest and Gradient Boosting were the models with the lowest mean squared error (MSE) overall, but both took a long time to train and were

sometimes overly confident. Both models fit the training data closely and this was concerning for overfitting. Linear Regression models were interpretable but were poor at picking up trends that frequently occur in non-linear data based on real housing data. This is a good example of the trade-off we face between accuracy and simplicity.

If there were more time, we would try to add more of the context behind the data, such as economic indicators, spatial distance to points of interest, or text from property listings to us for phrases like "water view" or "quiet neighbourhood". The next step in this process would be to deploy the models to a live tool that can be interactive for the agent-ca or client-facing through the Gradio model.